

## Microservices architecture

Containerisation often goes hand in hand with microservices architecture, where applications are divided into smaller, independently deployable services. In such an environment, monitoring is crucial for understanding the interactions between microservices, detecting bottlenecks, and ensuring that each service operates within its resource boundaries. Effective monitoring contributes to the overall stability and performance of the entire microservices-based application.

## Rapid scaling and orchestration

Containers facilitate rapid scaling to meet varying workloads. Container orchestration tools, such as Kubernetes and Docker Swarm, dynamically manage the deployment, scaling, and orchestration of containers. Monitoring tools integrated with these orchestrators provide insights into the automatic scaling behaviour, helping organisations understand how their applications respond to changing demand and ensuring that the infrastructure scales efficiently.

## Security and compliance

Monitoring is crucial for ensuring the security and compliance of containerised applications. Security-related metrics, such as network traffic, system calls, and container activity, can be monitored to detect and respond to potential security threats. Furthermore, compliance requirements often mandate the tracking of specific metrics, and monitoring tools help organisations demonstrate adherence to these standards.

## Cost management

Efficient monitoring and analysis contribute to better cost management in containerised environments. By understanding resource utilisation patterns and optimising container allocation, organisations can avoid unnecessary costs associated with

over-provisioning resources. This is particularly important in cloud-based environments where resource usage directly impacts expenses.

## Resource optimisation

Monitoring helps identify resource utilisation patterns, ensuring that containers have the necessary resources for optimal performance. This includes monitoring CPU usage, memory consumption, and storage utilisation. Containers share the host operating system's kernel, making them more efficient in terms of resource utilisation. However, this shared environment also necessitates careful monitoring to prevent resource contention. Monitoring tools help identify patterns in CPU usage, memory consumption, and storage utilisation, ensuring that each container has the necessary resources to perform optimally.

## Performance management

Containers can be ephemeral, with life cycles measured in seconds or minutes. Hence, performance issues can arise quickly. Monitoring tools enable real-time performance analysis, helping to identify and address bottlenecks promptly. By tracking metrics related to response times, throughput, and latency, organisations can proactively manage the performance of containerised applications and maintain a positive user experience.

## Scalability

Containers are designed to scale horizontally, and monitoring tools assist in dynamically adjusting resources to accommodate varying workloads.

## Fault detection and troubleshooting

Containerised environments are prone to faults, such as application failures, container crashes, or networking issues. Monitoring tools aid in the early detection of faults, triggering alerts or notifications when predefined thresholds are exceeded. Additionally, detailed

analytics provided by monitoring tools assist in troubleshooting, reducing the time it takes to identify and resolve issues, thereby minimising downtime.

## Free tools for monitoring and analytics of containers

Several free and open source tools are available to monitor and analyse containerised environments. These tools provide insights into container performance, resource utilisation, and overall system health. Here are some noteworthy options.

### cAdvisor (Container Advisor)

<https://github.com/google/cadvisor>  
cAdvisor, short for Container Advisor, stands out as a pivotal open source project developed by Google to cater to the burgeoning needs of containerised environments. With a focus on container-specific metrics, it serves as a robust solution for monitoring and managing containers seamlessly.

The primary objective of cAdvisor is to simplify the monitoring process by automating the collection, aggregation, processing, and exportation of essential information pertaining to running containers. Its capability to automatically gather data from containers contributes significantly to real-time insights into resource utilisation, performance metrics, and the overall health of containerised applications.

A distinguishing feature of cAdvisor is its versatility, as it supports a variety of container runtimes. Whether an organisation is utilising Docker, containerd, or other container runtimes, cAdvisor ensures compatibility, making it a versatile and widely applicable monitoring tool.



Figure 1: cAdvisor project